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MILITARY SPECIFICATION

CHEMICAL AGENT, PLASTICIZED WHITE PHOSPHOROUS (PWP)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of plasticized white phosphorus (PWP).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 — Sieves, Standard for Testing Purposes.

TT-X-916 — Xylene (for Use in Organic Coatings).

MILITARY

MIL-C-215 — Chemical Agent, White Phosphorus (WP).

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 791 — Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

FED. STD. No. 90 — Rubbers, Synthetic: Specification Limits, Sampling, and Testing.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FSC 1365

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CODE OF FEDERAL REGULATIONS

49 CFR 71–90 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–90.")

3. REQUIREMENTS

3.1 Composition. The PWP shall be compounded from a granulated white phosphorus—water slurry and plasticizing mixture in such manner that the resulting product will be a substantially uniform mixture of 75 ± 3 parts by weight of white phosphorus and 25 ± 3 parts by weight of plasticizing mixture.

3.1.1 *White phosphorus.* The white phosphorus granules shall conform to Specification MIL–C–215 and, when in the water slurry before mixing with the plasticizing mixture, shall be in such a state of subdivision that no less than 98 percent will pass a 4760–micron (No. 4) sieve and no less than 40 percent will pass a 590-micron (No. 30) sieve when tested as specified in 4.6.1.

3.1.2 *Plasticizing mixture.* The plasticizing mixture shall consist of 35 ± 1 parts by weight of synthetic rubber complying with the specification for GR–S–1000 contained in Federal Standard 90, and 65 ± 1 parts by weight of xylene conforming to Grade B of Specification TT–X–916.

3.2 Water content. The water content of the PWP shall not exceed 16 percent by weight when tested as specified in 4.6.2.

3.3 Thermal stability. The separation of white phosphorus from the PWP shall not exceed 5 percent when tested as specified in 4.6.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1, 3.1.2, and section 5 have been satisfied.

4.3 Lotting. A lot shall consist of the plasticized white phosphorus offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.4 Sampling.

4.4.1 *For nondestructive examination.* Sampling shall be conducted in accordance with Standard MIL–STD–105.

4.4.2 *For test.*

4.4.2.1 *White phosphorus—water slurry.* A sample of approximately ½ pint of white phosphorus slurry shall be taken from each batch at the time of transfer to the mixer. The sample shall be placed in a 1-pint sample jar and the jar sealed and labeled to identify the sample with the batch and lot represented. These batch samples shall be held pending completion of the lot. Three sample jars shall then be selected at random from each lot for use in the particle size determinations (see 4.6.1).

4.4.2.2 *Plasticized white phosphorus.* A sample of approximately 1 pint of PWP shall be taken at the extruder from each batch of PWP. The sample shall be placed in a 1-pint sample jar and the remaining air in the jar displaced with carbon dioxide. The jar shall be sealed and labeled to identify the sample with the batch and lot represented. These batch samples shall be held pending completion of the lot. Three sample jars shall then be selected at random from each lot for use in the water content (4.6.2) and thermal stability (4.6.3) test determinations.

4.5 Examination. Sample containers of PWP shall be examined for compliance with level A preparation for delivery requirements in accordance with the classification of defects and with Standard MIL–STD–105.

4.5.1 *Classification of defects.*

4.5.1.1 *Preparation for delivery (section 5).*

Categories	*Defects*
Critical —	None defined.
Major —	AQL 1.5 percent defective:
	101 — Containers incorrect.
	102 — Closure incorrect.
	103 — Containers damaged or leaking.
	104 — Nitrogen or water of incorrect quantity.
	105 — Marking and labeling incorrect, missing, or illegible.

4.6 Tests. Distilled water and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be made and corrections applied where significant. Tests shall be conducted as follows:

4.6.1 *Particle size.*

4.6.1.1 *White phosphorus granules weight determination.* Fill with tap water to the top graduation a graduated cylinder and weigh to the nearest 0.5 gram (g). empty its contents and fill to the top graduation with the white phosphorus slurry. Weigh to the nearest 0.5 g. Calculate the weight of solid white phosphorus in the graduate as follows:

$$\text{Grams of white phosphorus granules} = 1.83\ (B - A)$$

where: A = weight of graduate filled with water, in grams.

B = weight of graduate filled with the white phosphorus slurry, in grams.

4.6.1.2 *Procedure.* Transfer the entire contents of the sample jar into a 500-milliliter (ml.) graduated cylinder and determine the weight of the white phosphorus in accordance with the procedure given in 4.6.1.1. Select a 4760-micron (No. 4) and a 590-micron (No. 30) U.S. Standard sieve, conforming to Specification RR–S–366. nest them with the 4760-micron (No. 4) sieve on top, and place a loosely fitting pan on the bottom. Place the nest of sieves under a stream of water. Transfer the entire sample from the 500-ml. graduate into the top sieve. Move the sieves under the stream of tapwater until as many granules of white phosphorus

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as possible have been washed through the sieve. Keep granules wet at all time. Transfer the residue remaining on each sieve to a weighed 250-ml. graduated cylinder by means of a wash bottle and a powder funnel. Determine the weight of the residue on each sieve according to the procdure given in 4.6.1.1. Calculate the percent material passing through each sieve as follows:

$$\text{Percent passing through 4760-micron (No. 4) sieve} = \frac{100\ (W - A)}{W}$$

$$\text{Percent passing through 590-micron (No. 30) sieve} = \frac{100\ (W - A - B)}{W}$$

where: A = grams of residue on 4760-micron (No. 4) sieve.
B = grams of residue on 590-micron (No. 30) sieve, and
W = grams of white phosphorus in sample.

4.6.2 *Water content.* Determine the water content according to method 3001 of Federal Test Method Standard No. 791, using approximately 25 g of sample, and xylene, grade B of Specification TT–X–916, as the solvent.

4.6.3 *Thermal stability.* Weigh a 60-ml. large mouth, screw-top bottle and its cover to the nearest 0.1 g. Transfer approximately 20 to 30 g of the sample to the bottle, displace the remaining air wih carbon dioxide, cover, and weigh again to the nearest 0.1 g. Place in a constant temperature bath at 67° to 73° C. for 2 hours and then allow to cool to 20° to 30° C. If any of the white phosphorus has separated from the original mass of plasticized white phosphorus, remove the separated white phosphorus and determine by difference the weights of the white phosphorus and plasticized white phosphorus. Calculate the percent white phosphorus separated as follows:

$$\text{Percent white phosphorus separated} = \frac{100\ A}{W}$$

where: A = grams of white phosphorus separated, and
W = grams of sample used.

4.7 **Acceptance/rejection criteria.** If any sample fails to meet any requirement of the specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* Plasticized white phosphorus shall be packed in metal drums of 30-gallon nominal capacity with welded bottom and meeting the requirements of ICC Specification 6B. Immediately after the drum is filled, the remaining air in the drum shall be replaced with an inert blanket of nitrogen. Alternatively, when specified, the contents shall be blanketed with water, leaving sufficient outage in the drum to conform to ICC regulations.

5.1.2 *Level C.*

5.1.2.1 *Tank car quantities.* When speci-

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fied, plasticized white phosphorus shall be shipped in tank cars meeting the requirements of ICC Specification 103 or 103W. The remaining air in the tank car shall be replaced with an inert blanket of nitrogen. Alternatively, when specified, the contents shall be blanketed with water, leaving sufficient outage to conform to ICC regulations.

5.1.2.2 *Drums.* The plasticized white phosphorus shall be packed in drums in accordance with commercial manufacturing practice to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. The container used shall comply with ICC regulations or the common carrier regulations applicable to the mode of transportation.

5.2 **Marking.** In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with ICC regulations and Standard MIL–STD–129.

5.2.1 *Precautionary marking.* In addition to other markings required by the contract or order, the following precautionary markings shall be applied to each individual container.

"PLASTICIZED WHITE PHOSPHORUS
DANGER! CAUSES SEVERE BURNS.
Contents are packed under nitrogen or water and will ignite if nitrogen or water is removed.

Do not get on skin or in eyes.
Wear heavy rubber gloves and goggles or face shield. In case of contact, immediately flush skin or eyes with plenty of water for at least 15 minutes. Keep skin area wet until medical attention is obtained. Keep container out of sun and away from heat."

POISON



6. NOTES

6.1 **Intended use.** The material covered by this specification is intended for use in the manufacture of white smoke munitions.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packing required.

6.3 **Batch.** A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 **Significant places.** For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

6.5 **Caution.** White phosphorus is a highly flammable substance with an exceedingly low autoignition temperature. Every precaution should be taken during manufacture, sampling, storage, and testing to minimize danger to personnel and equipment.

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